

Chapter 7.6

Stowage and segregation on general cargo ships

7.6.1 Introduction

- 7.6.1.1 The provisions of this chapter apply to the stowage and segregation of dangerous goods stowed in the conventional way on board general cargo ships. They apply also to containers which are transported in conventional cargo spaces, including cargo spaces on the weather deck, not properly fitted to give a permanent stowage of the containers during transport.
- 7.6.1.2 For ships carrying containers in stowage positions which are properly fitted for the permanent stowage of containers the provisions of chapter 7.4 apply.

7.6.2 Stowage and handling provisions

7.6.2.1 Provisions for all classes

- 7.6.2.1.1 The minimum stacking height for testing packagings intended to contain dangerous goods in accordance with chapter 6.1 is 3 m. For IBCs and large packagings, the stacking test load shall be determined in accordance with 6.5.6.6.4 and 6.6.5.3.3.4 respectively.
- 7.6.2.1.2 Drums containing dangerous goods shall always be stowed in an upright position unless otherwise authorized by the competent authority.
- 7.6.2.1.3 The stowage of dangerous goods shall be so arranged as to ensure clear walkways and access to all facilities necessary for the safe working of the ship. When dangerous goods are stowed on deck, hydrants, sounding pipes and the like and access thereto shall be kept free and clear of such goods.
- 7.6.2.1.4 Fibreboard packagings, paper bags and other packages susceptible to water damage shall be stowed *under deck* or, if they are stowed on deck, they shall be so protected that at no time are they exposed to weather or to seawater.
- 7.6.2.1.5 Portable tanks shall not be overstowed by other cargo unless they are designed for that purpose or unless they are protected to the satisfaction of the competent authority.
- 7.6.2.1.6 Cargo spaces and decks shall be clean and dry as relevant to the hazards of the dangerous goods to be carried. In order to reduce the risk of ignition, the space shall be free of dust from other cargoes, such as grain or coal dust.
- 7.6.2.1.7 Packages and cargo transport units found to be damaged, leaking or sifting shall not be loaded on a general cargo ship. Care shall be taken to ensure that excessive water, snow, ice or foreign matter adhering to packages and cargo transport units shall be removed before loading.
- 7.6.2.1.8 Packages and cargo transport units and any other goods shall be adequately braced and secured for the voyage.* Packages shall be loaded in such a way that there will be a minimum likelihood of damage to them and to any fittings during transport. Fittings on packages or portable tanks shall be adequately protected.

7.6.2.2 Provisions for flammable gases and highly flammable liquids

- 7.6.2.2.1 In cargo ships of 500 gross tons or over and passenger ships constructed before 1 September 1984, and in cargo ships of less than 500 gross tons constructed before 1 February 1992, flammable gases or flammable liquids having a flashpoint of less than 23°C c.c., shall be stowed on deck only, unless otherwise approved by the Administration.
- 7.6.2.2.2 Flammable gases or liquids having a flashpoint less than 23°C c.c. transported on deck shall be stowed at least 3 m from any potential source of ignition.

* Refer to regulation VII/5 of SOLAS, as amended.

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7.6.2.3 Ventilation provisions

7.6.2.3.1 In cargo ships of 500 gross tons or over and passenger ships constructed before 1 September 1984, and in cargo ships of less than 500 gross tons constructed before 1 February 1992, the following dangerous goods, may be stowed *under deck* only if the cargo space is equipped with mechanical ventilation and if under deck stowage is permitted in the Dangerous Goods List:

- dangerous goods of class 2.1;
- dangerous goods of class 3 with a flashpoint of less than 23°C c.c.;
- dangerous goods of class 4.3;
- dangerous goods of class 6.1 with a subsidiary hazard of class 3;
- dangerous goods of class 8 with a subsidiary hazard of class 3, and
- dangerous goods to which a specific stowage requirement requiring mechanical ventilation in column 16a of the Dangerous Goods List is assigned.

Otherwise containers shall be stowed on deck only.

7.6.2.3.2 The capacity of the mechanical ventilation (number of air changes per hour) shall be to the satisfaction of the Administration.

7.6.2.4 Provisions for class 1

7.6.2.4.1 All compartments or holds and cargo transport units shall be locked or suitably secured in order to prevent unauthorized access. The means of locking and securing shall be such that, in the case of emergency, access can be gained without delay.

7.6.2.4.2 Loading and unloading procedures and equipment used should be of such a nature that sparks are not produced, in particular where the floors of the cargo compartment are not constructed of close-boarded wood. All cargo handlers should be briefed by the shipper or receiver of the potential risks and necessary precautions, prior to commencing the handling of explosives. In the event of the contents of packages being affected by water whilst on board, immediate advice shall be sought from the shipper; pending this advice, handling of the packages shall be avoided.

7.6.2.4.3 Segregation on deck

When goods in different compatibility groups are transported on deck, they shall be stowed not less than 6 m apart unless their mixed stowage is allowed according to 7.2.7.

7.6.2.4.4 Segregation in single hold ships

In a single hold ship, dangerous goods of class 1 shall be segregated in accordance with 7.2.7 except that:

- .1 Goods in Division 1.1 or 1.2 of compatibility group B may be stowed in the same hold as substances of compatibility group D provided:
 - the net explosives mass of goods of compatibility group B does not exceed 50 kg; and
 - such goods are stowed in a closed cargo transport unit which is stowed at least 6 m from the substances of compatibility group D.
- .2 Goods in Division 1.4 of compatibility group B may be stowed in the same hold as substances of compatibility group D provided they are separated either by a distance of at least 6 m or by a steel division.

7.6.2.4.5 In the event that a package containing goods of class 1 is found to be suffering from breakage or leakage expert advice should be obtained for its safe handling and disposal.

7.6.2.5 Provisions for class 2

7.6.2.5.1 When pressure receptacles are stowed in a vertical position they shall be stowed in a block, cribbed or boxed-in with suitable sound lumber and the box or crib dunnaged to provide clearance from a steel deck. Pressure receptacles in a box or crib shall be braced to prevent any movement. The box or crib (gas rack) shall be securely chocked and lashed to prevent movement in any direction.

7.6.2.5.2 Pressure receptacles stowed on deck shall be protected from sources of heat.

7.6.2.6 Provisions for class 3

7.6.2.6.1 Class 3 substances with a flashpoint of less than 23°C c.c. packaged in jerricans, plastics (3H1, 3H2), drums, plastics (1H1, 1H2), plastics receptacles in a plastic drum (6HH1, 6HH2) and Plastic Intermediate Bulk Containers (IBCs 31H1 and 31H2), shall be stowed *on deck only* unless packed in a closed cargo transport unit.

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7.6.2.6.2 Packages loaded on deck shall be protected from sources of heat.

7.6.2.7 Provisions for classes 4.1, 4.2 and 4.3

7.6.2.7.1 Packages stowed on deck shall be protected from sources of heat.

△ 7.6.2.7.2 Stowage provisions for CARBON, animal or vegetable origin (UN 1361), FISH MEAL, UNSTABILIZED (UN 1374), FISH MEAL, STABILIZED (UN 2216, class 9) and KRILL MEAL (UN 3497)

7.6.2.7.2.1 For loose packages:

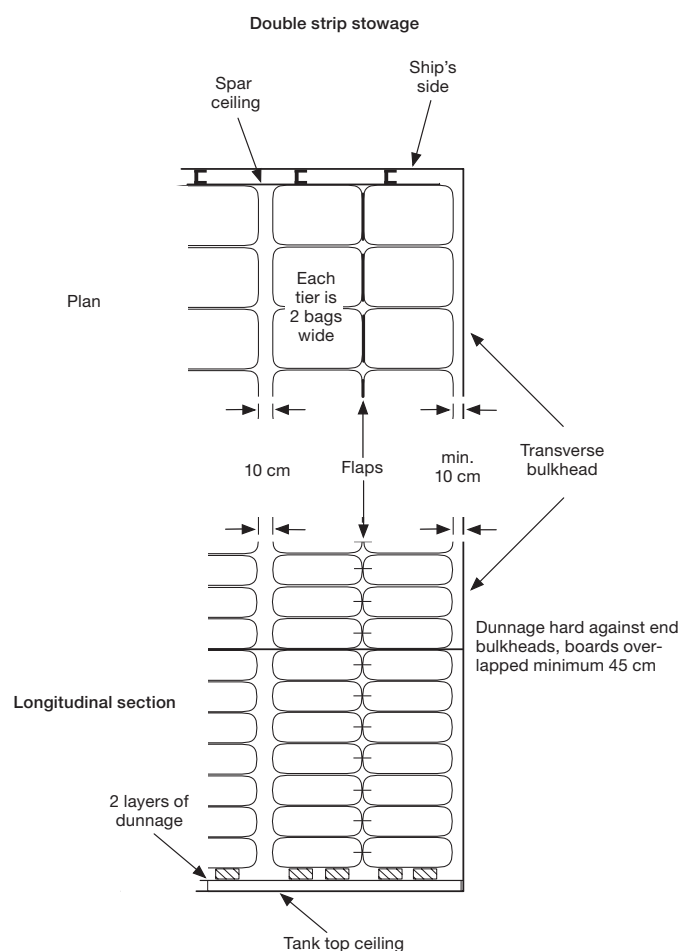
- .1 Temperature readings shall be taken 3 times a day during the voyage and recorded.
- .2 If the temperature of the cargo exceeds 55°C and continues to increase, ventilation to the hold shall be restricted. If self-heating continues, then carbon dioxide or inert gas shall be introduced. The ship shall be equipped with facilities for introducing carbon dioxide or inert gas into the holds.
- .3 The cargo shall be stowed protected from sources of heat.

△ .4 For UN 1361, 1374 and 3497, where loose bags are being carried, double strip stowage is recommended, provided there is good surface and through ventilation. The diagram in 7.6.2.7.2.3 shows how this can be achieved. For UN 2216, where loose bags are being carried, no special ventilation is required for block stowage of bagged cargo.

7.6.2.7.2.2 For containers:

- .1 After packing, the doors and other openings shall be sealed to prevent the penetration of air into the unit.
- .2 Temperature readings in the hold shall be taken once a day early in the morning during the voyage and recorded.
- .3 If the temperature of the hold rises excessively above ambient and continues to increase, the possible need to apply copious quantities of water in an emergency and the consequent risk to the stability of the ship shall be considered.
- .4 The cargo shall be stowed protected from sources of heat.

7.6.2.7.2.3



7.6.2.7.3 Stowage provisions for SEED CAKE (UN 1386)

7.6.2.7.3.1 Stowage provisions for SEED CAKE, containing vegetable oil (a) mechanically expelled seeds, containing more than 10% oil or more than 20% oil and moisture combined:

- .1 through and surface ventilation is required;
- .2 if the voyage exceeds 5 days, the ship shall be equipped with facilities for introducing carbon dioxide or inert gas into the cargo spaces;
- .3 bags shall always be stowed in double strip, as shown in 7.6.2.7.2.3 of this Code for fish meal, unstabilized; and
- .4 regular temperature readings shall be taken at varying depths in the cargo space and recorded. If the temperature of the cargo exceeds 55°C and continues to increase, ventilation to the cargo spaces shall be restricted. If self-heating continues, then carbon dioxide or inert gas shall be introduced.

7.6.2.7.3.2 Stowage provisions for SEED CAKE, containing vegetable oil (b) solvent extractions and expelled seeds containing not more than 10% of oil and, when the amount of moisture is higher than 10%, not more than 20% of oil and moisture combined:

- .1 surface ventilation is required to assist in removing any residual solvent vapour;
- .2 if bags are stowed without provision for ventilation to circulate throughout the stow and the voyage exceeds 5 days, regular temperature readings shall be taken at varying depths in the hold and recorded; and
- .3 if the voyage exceeds 5 days, the vessel shall be equipped with facilities for introducing carbon dioxide or inert gas into the cargo spaces.

7.6.2.8 Provisions for class 5.1

7.6.2.8.1 Cargo spaces shall be cleaned before oxidizing substances are loaded into them. All combustible materials which are not necessary for the stowage of such cargoes shall be removed from the hold.

7.6.2.8.2 As far as reasonably practicable, non-combustible securing and protecting materials and only a minimum of clean dry wooden dunnage shall be used.

7.6.2.8.3 Precautions shall be taken to avoid the penetration of oxidizing substances into other cargo spaces, bilges, etc., which may contain combustible material.

△ **7.6.2.8.4** UN 1942 AMMONIUM NITRATE and UN 2067 AMMONIUM NITRATE BASED FERTILIZER may be stowed under deck in a clean cargo space capable of being opened up in an emergency. The possible need to open hatches in case of fire to provide maximum ventilation and to apply water in an emergency and the consequent risk to the stability of the ship through flooding of cargo space shall be considered before loading. The requirement for opening the cargo space hatches applies to the weather deck and tween deck hatches (if any).

7.6.2.8.5 After discharge, cargo spaces used for the transport of oxidizing substances shall be inspected for contamination. A space that has been contaminated shall be properly cleaned and examined before being used for other cargoes.

7.6.2.9 Provisions for self-reactive substances of class 4.1 and for class 5.2

7.6.2.9.1 Packages shall be stowed protected from sources of heat.

7.6.2.9.2 When stowage arrangements are made, it shall be borne in mind that it may become appropriate to jettison a package or packages of this cargo.

7.6.2.10 Provisions for classes 6.1 and 8

7.6.2.10.1 After discharge, spaces used for the transport of substances of this class shall be inspected for contamination. A space which has been contaminated shall be properly cleaned and examined before being used for other cargoes.

7.6.2.10.2 Substances of class 8 shall be kept as dry as reasonably practicable, since in the presence of moisture they may be corrosive to most metals and some also react violently with water.

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7.6.2.11 Stowage of goods of class 9**7.6.2.11.1 Stowage provisions for AMMONIUM NITRATE BASED FERTILIZER, UN 2071**

7.6.2.11.1.1 AMMONIUM NITRATE BASED FERTILIZER, UN 2071 shall be stowed in a clean cargo space capable of being opened up in an emergency. In the case of bagged fertilizer or fertilizer in containers or in bulk containers, it is sufficient if, in the case of an emergency, the cargo is accessible through free approaches (hatch entries), and mechanical ventilation enables the master to exhaust any gases or fumes resulting from decomposition. The possible need to open hatches in case of fire to provide maximum ventilation and to apply water in an emergency, and the consequent risk to the stability of the ship through flooding of the cargo space, shall be considered before loading.

7.6.2.11.1.2 If suppression of decomposition should prove impracticable (such as in bad weather), there would not necessarily be immediate danger to the structure of the ship. However, the residue left after decomposition may have only half the mass of the original cargo; this loss of mass may also affect the stability of the ship and shall be considered before loading.

7.6.2.11.1.3 AMMONIUM NITRATE BASED FERTILIZER, UN 2071 shall be stowed out of direct contact with a metal engine-room bulkhead. In the case of bagged material, this may be done, for example, by using wooden boards to provide an air space between the bulkhead and the cargo. This requirement need not apply to short international voyages.

7.6.2.11.1.4 In the case of ships not fitted with smoke-detecting or other suitable devices, arrangements shall be made during the voyage to inspect cargo spaces containing these fertilizers at intervals not exceeding 4 h (such as to sniff at the ventilators serving them) to ensure early detection of decomposition should that occur.

7.6.2.11.2 Stowage provisions for FISH MEAL, STABILIZED (UN 2216, class 9)

7.6.2.11.2.1 For stowage provisions for FISH MEAL, STABILIZED (UN 2216, class 9), see 7.6.2.7.2.

7.6.2.12 Stowage of dangerous goods in flexible bulk containers

7.6.2.12.1 The stowage of dangerous goods in flexible bulk containers is not permitted on deck.

7.6.2.12.2 Flexible bulk containers shall be stowed in such a way that there are no void spaces between flexible bulk containers in the hold. If the flexible bulk containers do not completely fill the hold, adequate measures shall be taken to avoid shifting of cargo.

7.6.2.12.3 The maximum permissible stacking height of flexible bulk containers shall never exceed three high.

7.6.2.12.4 When flexible bulk containers are fitted with venting devices, the stowage of the flexible bulk containers shall not impede their function.

7.6.3 Segregation provisions**7.6.3.1 Segregation from foodstuffs**

7.6.3.1.1 For the purpose of this subsection, the terms “away from”, “separated from” and “separated by a complete compartment or hold from” are defined in 7.6.3.2.

7.6.3.1.2 Dangerous goods having a primary or subsidiary hazard of classes 2.3, 6.1, 7 (with the exception of UN 2908, 2909, 2910 and 2911), 8 and dangerous goods for which it is referred to in segregation code SG29 or SG50 in column 16b of the Dangerous Goods List stowed in a conventional way shall be “separated from” foodstuffs stowed in a conventional way. If either dangerous goods or foodstuffs are in a closed cargo transport unit, dangerous goods shall be stowed “away from” foodstuffs. If both dangerous goods and foodstuffs are in different closed cargo transport units, no segregation requirements shall apply.

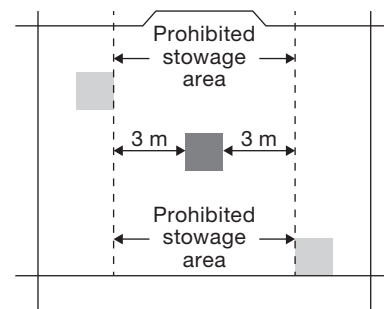
7.6.3.1.3 Dangerous goods of class 6.2 stowed in a conventional way shall be “separated by a complete compartment or hold from” foodstuffs stowed in a conventional way. If either dangerous goods or foodstuffs are in a closed cargo transport unit, dangerous goods shall be stowed “separated from” foodstuffs.

7.6.3.2 Segregation of packages containing dangerous goods and stowed in the conventional way

Definitions of the segregation terms

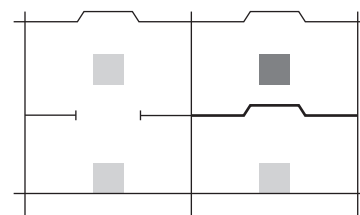
Away from:

Effectively segregated so that the incompatible goods cannot interact dangerously in the event of an accident but may be transported in the same compartment or hold or *on deck*, provided a minimum horizontal separation of 3 m, projected vertically, is obtained.



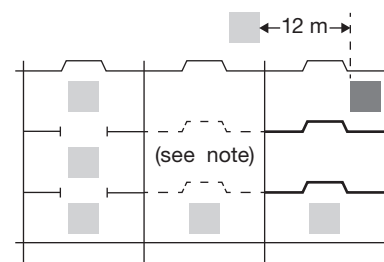
Separated from:

In different compartments or holds when stowed *under deck*. Provided the intervening deck is resistant to fire and liquid, a vertical separation, i.e. in different compartments, may be accepted as equivalent to this segregation. For *on deck* stowage, this segregation means a separation by a distance of **at least 6 m horizontally**.



Separated by a complete compartment or hold from:

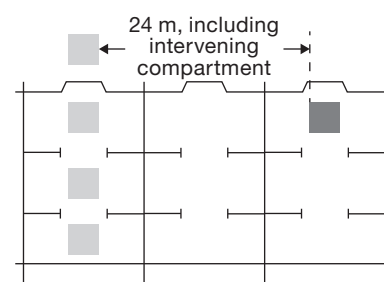
Either a vertical or a horizontal separation. If the intervening decks are not resistant to fire and liquid, then only a longitudinal separation, i.e. by an intervening complete compartment or hold, is acceptable. For *on deck* stowage, this segregation means a separation by a distance of **at least 12 m horizontally**. The same distance has to be applied if one package is stowed on deck, and the other one in an upper compartment.



Note: One of the two decks must be resistant to fire and to liquid

Separated longitudinally by an intervening complete compartment or hold from:

Vertical separation alone does not meet this requirement. Between a package *under deck* and one on deck, a minimum distance of 24 m, including a complete compartment, must be maintained longitudinally. For *on deck* stowage, this segregation means a separation by a distance of **at least 24 m longitudinally**.



Legend

Reference package	
Package containing incompatible goods	
Deck resistant to fire and liquid	

Note: Vertical lines represent transverse watertight bulkheads between cargo spaces.

Part 7 – Provisions concerning transport operations**7.6.3.3 Segregation of dangerous goods stowed in the conventional way from those transported in cargo transport units**

7.6.3.3.1 Dangerous goods stowed in the conventional way shall be segregated from goods transported in open cargo transport units in accordance with 7.6.3.2.

7.6.3.3.2 Dangerous goods stowed in the conventional way shall be segregated from goods transported in closed cargo transport units in accordance with 7.6.3.2 except that:

- .1 where “away from” is required, no segregation between the packages and the closed cargo transport units is required; and
- .2 where “separated from” is required, the segregation between the packages and the closed cargo transport units may be as for “away from” as defined in 7.6.3.2.

7.6.3.4 Segregation of dangerous goods in cargo transport units stowed in conventional cargo spaces

7.6.3.4.1 Dangerous goods in different closed cargo transport units (closed freight containers) stowed in holds and compartments not properly fitted to give a permanent stowage of the containers during transport shall be segregated from each other in accordance with 7.6.3.2 except that:

- .1 where “away from” is required, no segregation between the closed cargo transport units is required; and
- .2 where “separated from” is required, the segregation between the closed cargo transport units may be as for “away from” as defined in 7.6.3.2.

7.6.3.5 Segregation between bulk materials possessing chemical hazards and dangerous goods in packaged form

7.6.3.5.1 Unless otherwise required in this Code or in the IMSBC Code, segregation between bulk materials possessing chemical hazards and dangerous goods in packaged form shall be in accordance with the following table.

7.6.3.5.2 Segregation table

Bulk materials (classified as dangerous goods)	Dangerous goods in packaged form																
	CLASS	1.1 1.2 1.5	1.3 1.6	1.4	2.1	2.2 2.3	3	4.1	4.2	4.3	5.1	5.2	6.1	6.2	7	8	9
Flammable solids	4.1	4	3	2	2	2	2	X	1	X	1	2	X	3	2	1	X
Substances liable to spontaneous combustion	4.2	4	3	2	2	2	2	1	X	1	2	2	1	3	2	1	X
Substances which, in contact with water, emit flammable gases	4.3	4	4	2	2	X	2	X	1	X	2	2	X	2	2	1	X
Oxidizing substances (agents)	5.1	4	4	2	2	X	2	1	2	2	X	2	1	3	1	2	X
Toxic substances	6.1	2	2	X	X	X	X	X	1	X	1	1	X	1	X	X	X
Radioactive material	7	2	2	2	2	2	2	2	2	2	1	2	X	3	X	2	X
Corrosive substance	8	4	2	2	1	X	1	1	1	1	2	2	X	3	2	X	X
Miscellaneous dangerous substances and articles	9	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Materials hazardous only in bulk (MHB)		X	X	X	X	X	X	X	X	X	X	X	X	3	X	X	X

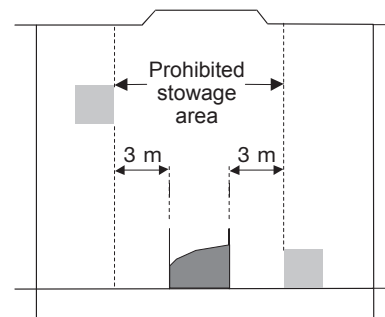
Numbers and symbols relate to the following terms, as defined in this chapter:

- 1 – “away from”
- 2 – “separated from”
- 3 – “separated by a complete compartment or hold from”
- 4 – “separated longitudinally by an intervening complete compartment or hold from”
- X – the segregation, if any, is shown in the Dangerous Goods List in this Code or the individual entries in the IMSBC Code

7.6.3.5.3 Definitions of the segregation terms

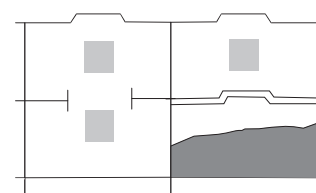
Away from:

Effectively segregated so that incompatible materials cannot interact dangerously in the event of an accident but may be transported in the same compartment or hold or *on deck* provided a minimum horizontal separation of 3 m, projected vertically, is provided.



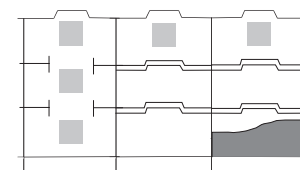
Separated from:

In different holds when stowed *under deck*. Provided an intervening deck is resistant to fire and liquid, a vertical separation, i.e. in different compartments, may be accepted as equivalent to this segregation.



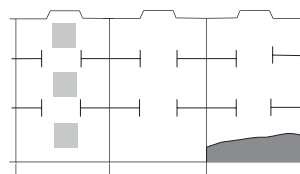
Separated by a complete compartment or hold from:

Either a vertical or a horizontal separation. If the decks are not resistant to fire and liquid, then only a longitudinal separation, i.e. by an intervening complete compartment, is acceptable.



Separated longitudinally by an intervening complete compartment or hold from:

Vertical separation alone does not meet this requirement.



Legend

- Reference bulk material 
- Package containing incompatible goods 
- Deck resistant to fire and liquid 

Note: Vertical lines represent transverse watertight bulkheads between cargo spaces.